

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: df = pd.read_csv('Telco-Customer-Churn.csv')
df.head()
```

```
Out[3]:
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	Dev
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	

5 rows × 21 columns



```
In [4]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype  
---  --
 0   customerID            7043 non-null   object  
 1   gender                7043 non-null   object  
 2   SeniorCitizen         7043 non-null   int64   
 3   Partner               7043 non-null   object  
 4   Dependents            7043 non-null   object  
 5   tenure                7043 non-null   int64   
 6   PhoneService          7043 non-null   object  
 7   MultipleLines         7043 non-null   object  
 8   InternetService       7043 non-null   object  
 9   OnlineSecurity        7043 non-null   object  
10   OnlineBackup          7043 non-null   object  
11   DeviceProtection      7043 non-null   object  
12   TechSupport           7043 non-null   object  
13   StreamingTV           7043 non-null   object  
14   StreamingMovies       7043 non-null   object  
15   Contract              7043 non-null   object  
16   PaperlessBilling      7043 non-null   object  
17   PaymentMethod         7043 non-null   object  
18   MonthlyCharges        7043 non-null   float64  
19   TotalCharges          7043 non-null   object  
20   Churn                 7043 non-null   object  
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

```

```

In [5]: #replacing blanks with 0 as tenure is 0 and no total charges recorded
df["TotalCharges"]=df["TotalCharges"].replace(" ", "0")
df["TotalCharges"]=df["TotalCharges"].astype(float)

```

```

In [6]: df.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
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 0   customerID            7043 non-null   object 
 1   gender                7043 non-null   object 
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 3   Partner               7043 non-null   object 
 4   Dependents            7043 non-null   object 
 5   tenure                7043 non-null   int64  
 6   PhoneService          7043 non-null   object 
 7   MultipleLines         7043 non-null   object 
 8   InternetService       7043 non-null   object 
 9   OnlineSecurity        7043 non-null   object 
10   OnlineBackup          7043 non-null   object 
11   DeviceProtection      7043 non-null   object 
12   TechSupport           7043 non-null   object 
13   StreamingTV           7043 non-null   object 
14   StreamingMovies       7043 non-null   object 
15   Contract              7043 non-null   object 
16   PaperlessBilling      7043 non-null   object 
17   PaymentMethod         7043 non-null   object 
18   MonthlyCharges        7043 non-null   float64 
19   TotalCharges          7043 non-null   float64 
20   Churn                 7043 non-null   object 
dtypes: float64(2), int64(2), object(17)
memory usage: 1.1+ MB

```

```

In [7]: #checking null values
df.isnull().sum().sum()

```

```

Out[7]: np.int64(0)

```

```

In [8]: #describing data
df.describe()

```

Out[8]:

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

In [9]: `df["customerID"].duplicated().sum()`Out[9]: `np.int64(0)`

```
In [10]: #function for replcationg int to txt
def conv(value):
    if value==1:
        return "yes"
    else:
        return "no"
df["SeniorCitizen"]=df["SeniorCitizen"].apply(conv)
```

In [11]: `df.head()`

Out[11]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	Dev
0	7590-VHVEG	Female	no	Yes	No	1	No	No phone service	DSL	No	...	
1	5575-GNVDE	Male	no	No	No	34	Yes	No	DSL	Yes	...	
2	3668-QPYBK	Male	no	No	No	2	Yes	No	DSL	Yes	...	
3	7795-CFOCW	Male	no	No	No	45	No	No phone service	DSL	Yes	...	
4	9237-HQITU	Female	no	No	No	2	Yes	No	Fiber optic	No	...	

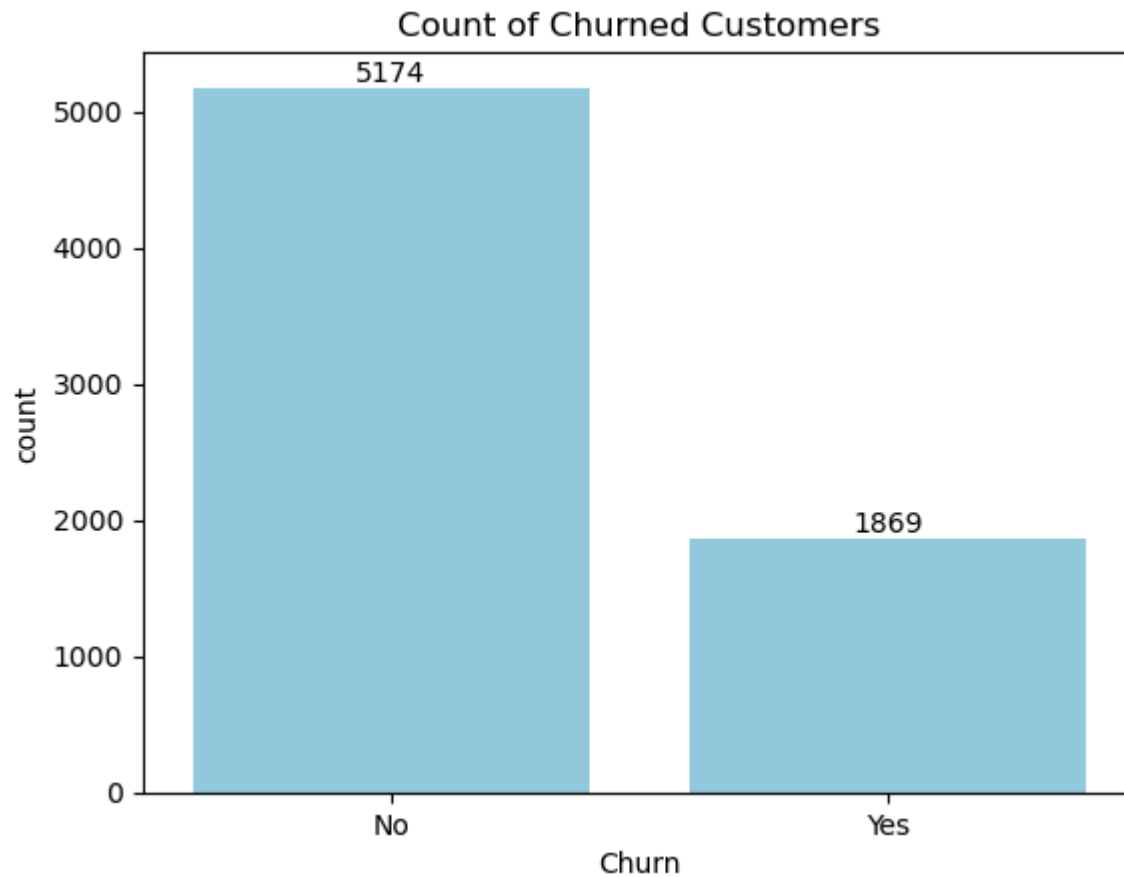
5 rows × 21 columns



```

In [12]: #how many customers have churned out
ax=sns.countplot(x='Churn',data=df, color="skyblue")
ax.bar_label(ax.containers[0])
plt.title("Count of Churned Customers")
plt.show()

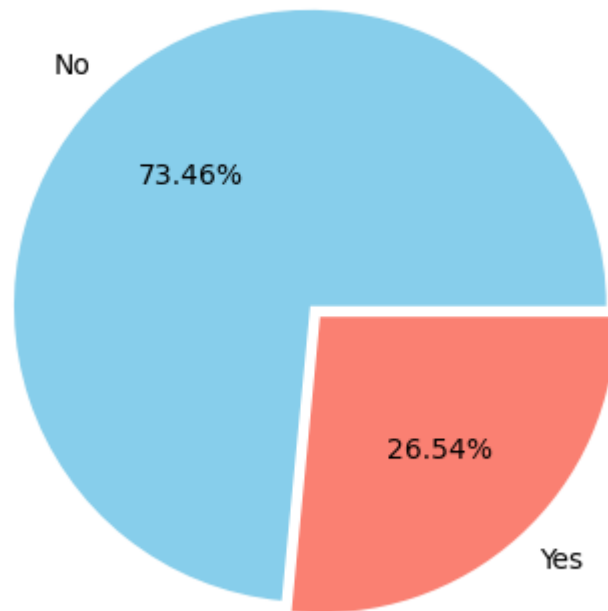
```



Most customers stayed with the company, but a significant number still left, indicating room to improve customer retention.

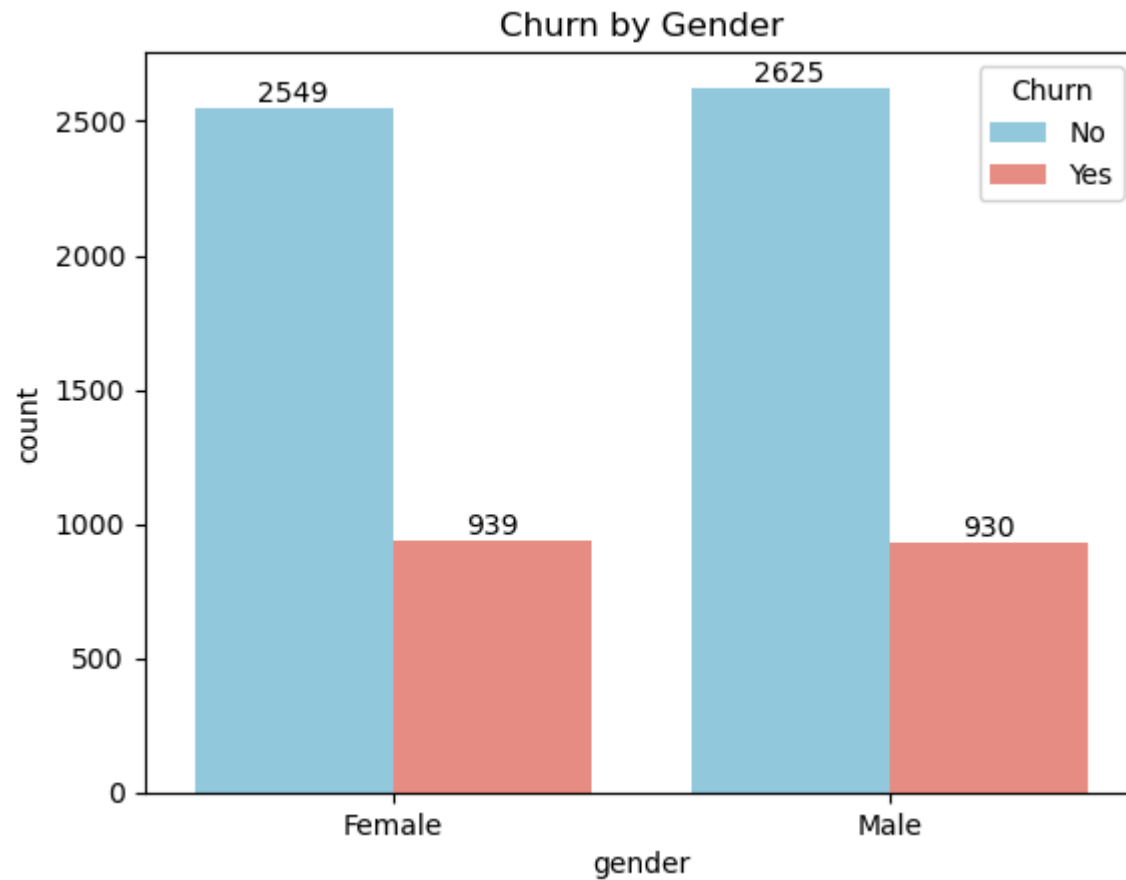
```
In [22]: #in percentage wise
gb=df.groupby('Churn').agg({'Churn':"count"})
plt.pie(gb['Churn'],labels=gb.index,autopct='%1.2f%%',explode=[0.05, 0],colors=["skyblue","salmon"])
plt.title("Percentage of Churned Customers")
plt.show()
```

Percentage of Churned Customers



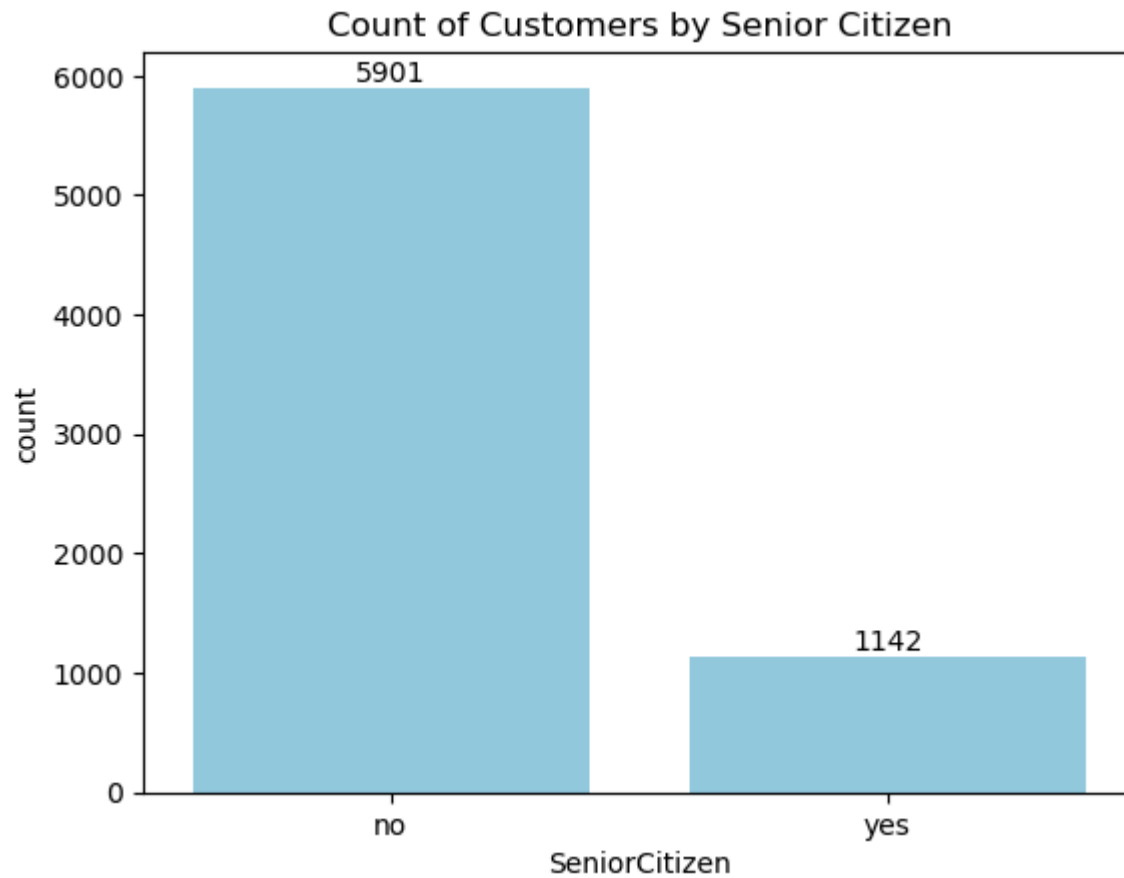
Around one-quarter of customers churned, meaning nearly 1 in 4 customers stopped using the service.

```
In [14]: ax=sns.countplot(x="gender",data=df,hue="Churn",palette=["skyblue", "salmon"])
plt.bar_label(ax.containers[0])
plt.bar_label(ax.containers[1])
plt.title("Churn by Gender",fontsize=12)
plt.show()
```



Male and female customers show very similar churn behavior, so gender has little impact on customer retention.

```
In [15]: ax=sns.countplot(x="SeniorCitizen",data=df, color="skyblue")
ax.bar_label(ax.containers[0])
plt.title("Count of Customers by Senior Citizen",fontsize=12)
plt.show()
```

The customer base mainly consists of non-senior citizens.

```
In [16]: ct = pd.crosstab(df["SeniorCitizen"], df["Churn"], normalize="index") * 100

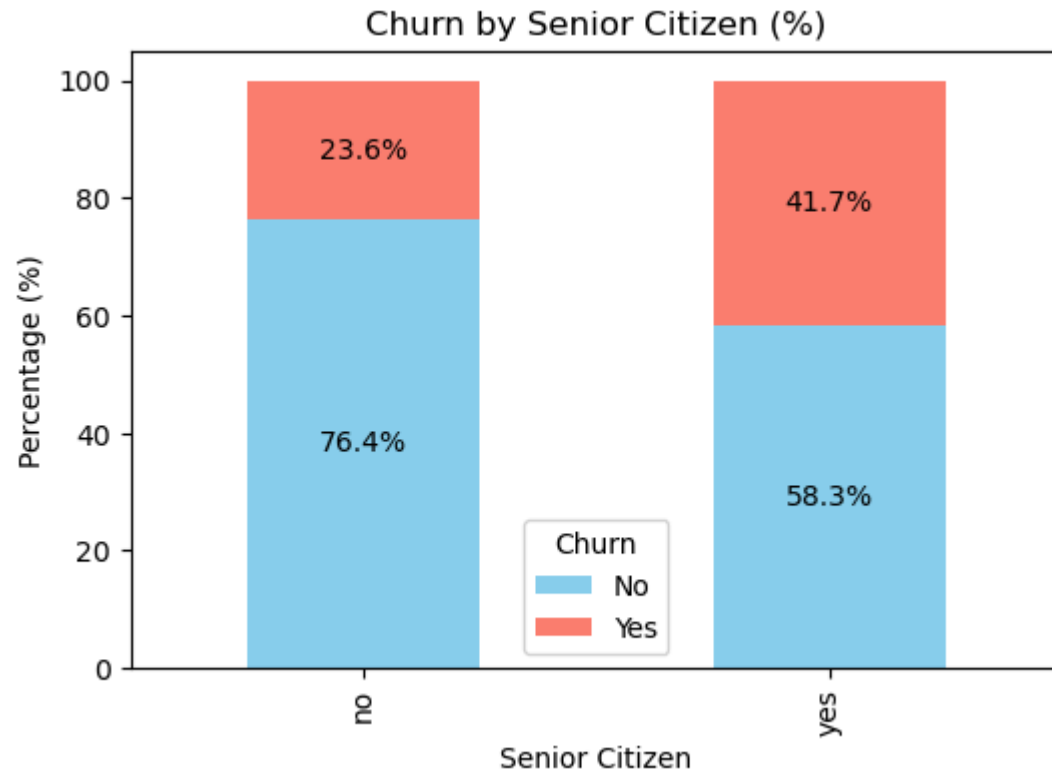
# Plot
ct.plot(kind="bar",
        stacked=True,
        figsize=(6,4),
        color=["skyblue", "salmon"])

plt.title("Churn by Senior Citizen (%)")
plt.xlabel("Senior Citizen")
```

```
plt.ylabel("Percentage (%)")
plt.legend(title="Churn")

# Add percentage labels
for c in plt.gca().containers:
    plt.gca().bar_label(c, fmt='%.1f%', label_type='center', color='black')

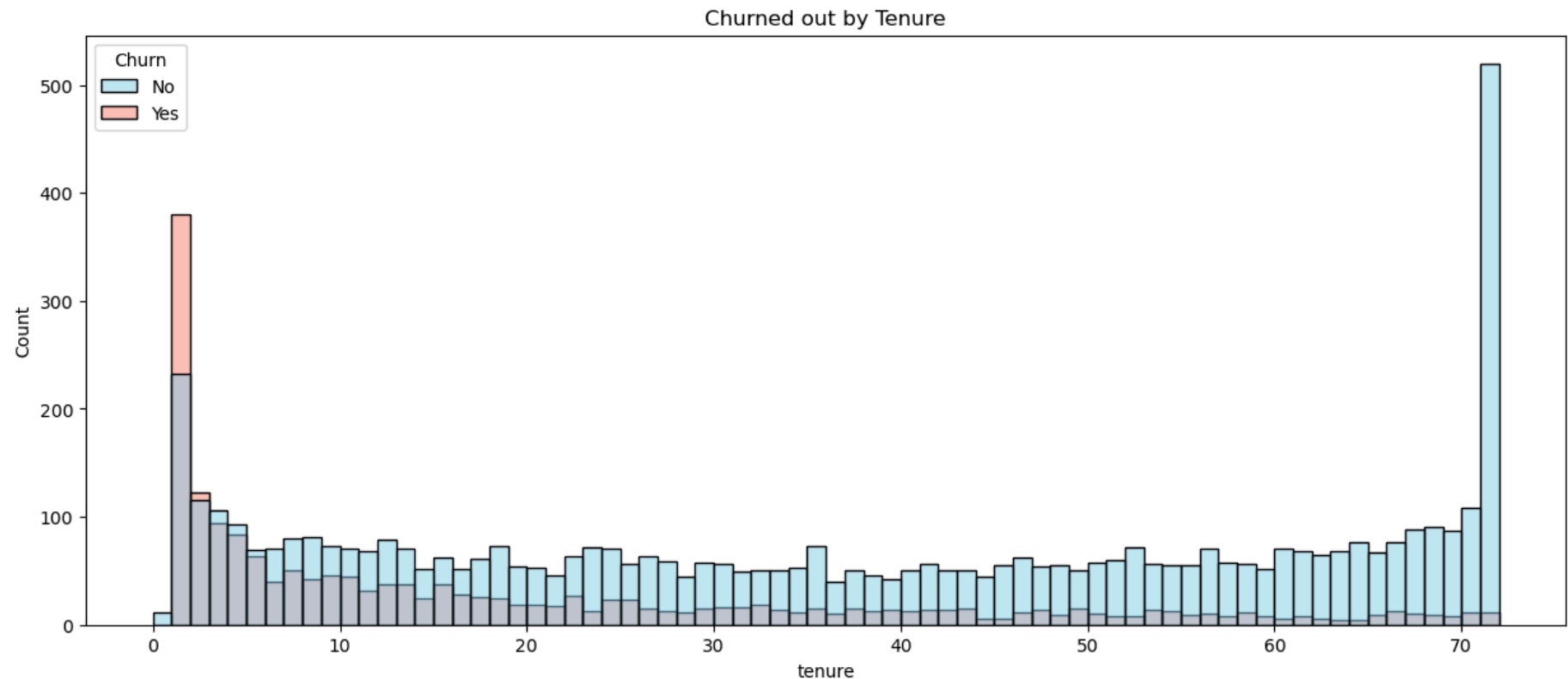
plt.show()
```



Senior citizens have a higher churn rate than non-senior customers, making them a higher-risk customer segment.

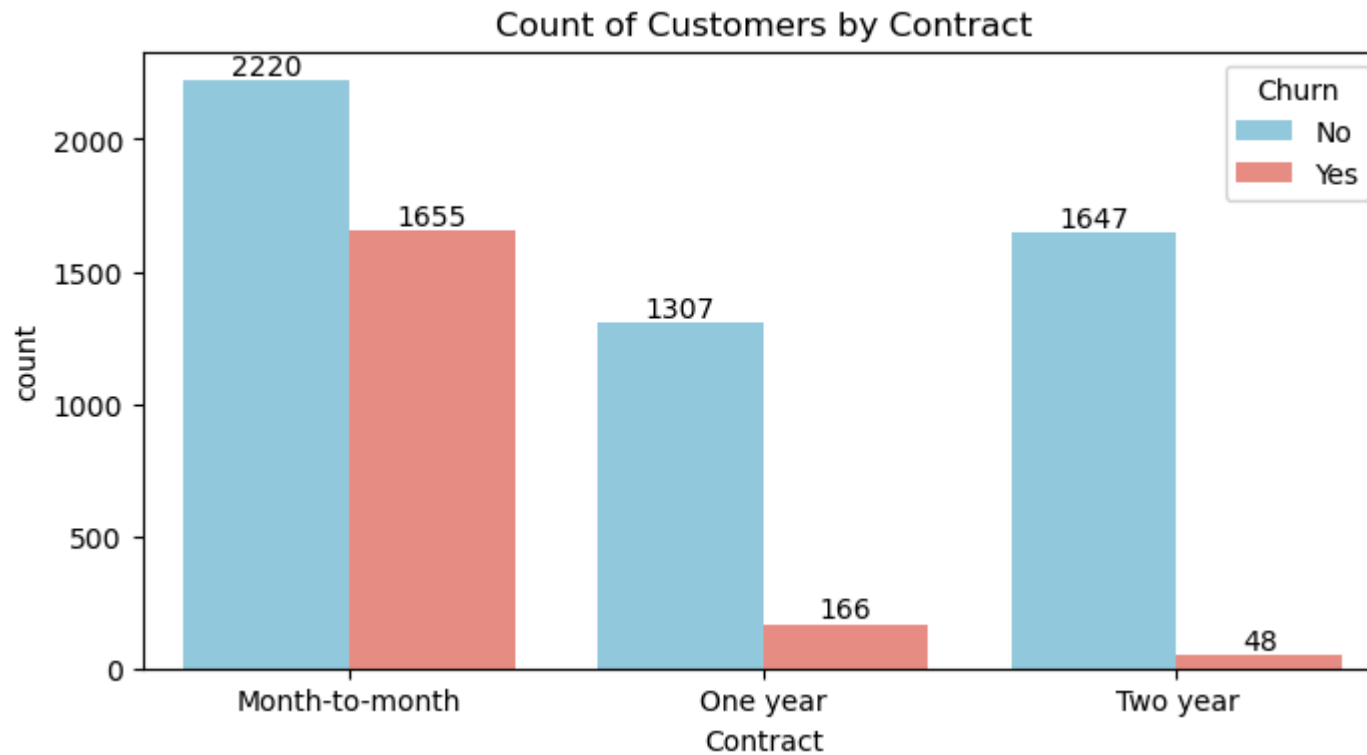
```
In [17]: plt.figure(figsize=(15,6))
sns.histplot(x="tenure", data=df, bins=72, hue="Churn", palette=["skyblue", "salmon"])
```

```
plt.title("Churned out by Tenure")
plt.show()
```



Most customers who churn leave within the early months, while long-term customers are more likely to stay.

```
In [18]: plt.figure(figsize=(8,4))
ax=sns.countplot(x="Contract",data=df,hue="Churn",palette=["skyblue", "salmon"])
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
plt.title("Count of Customers by Contract",fontsize=12)
plt.show()
```



Customers with month-to-month contracts churn much more than those with longer-term contracts.

```
In [19]: df.columns.values
```

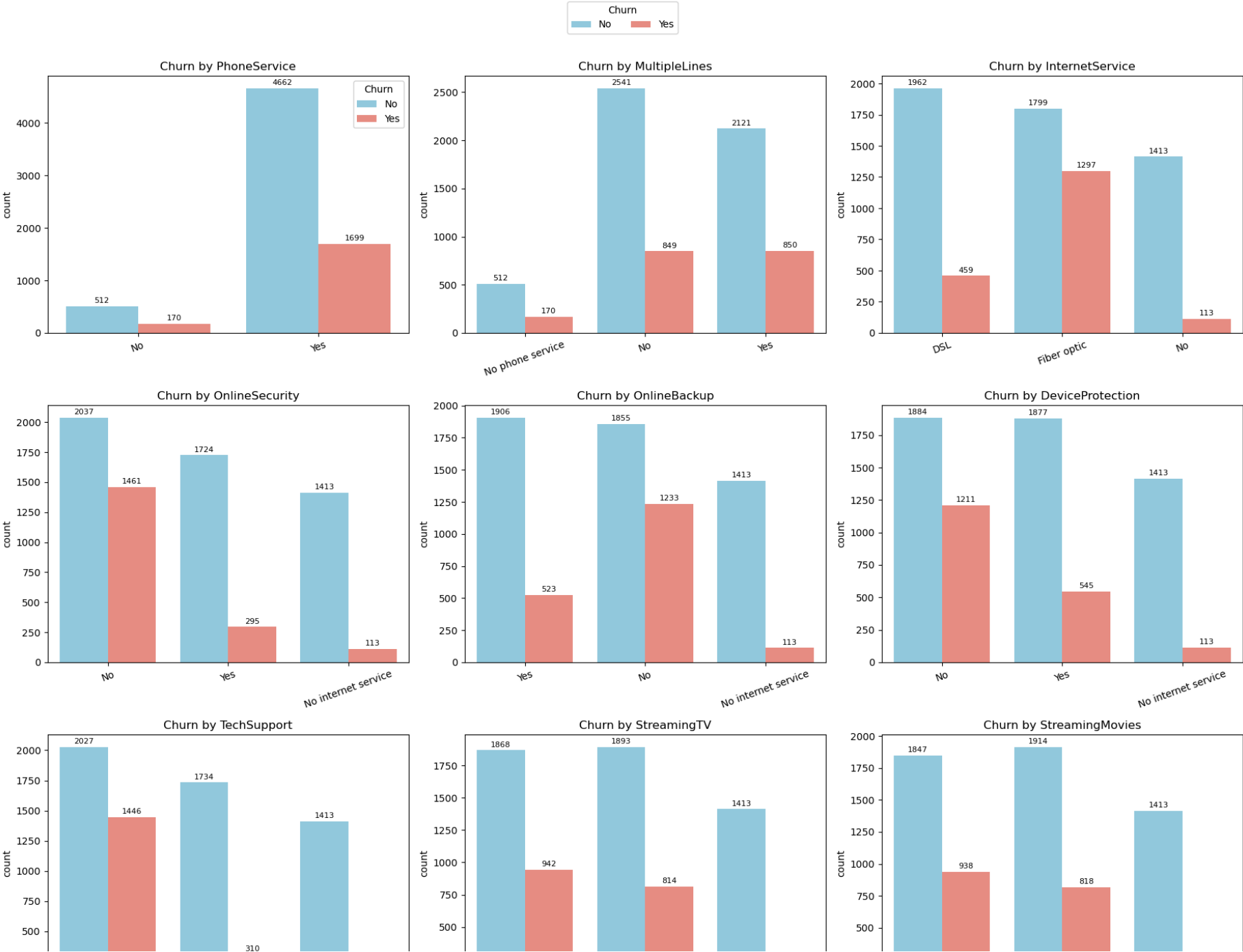
```
Out[19]: array(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',  
              'tenure', 'PhoneService', 'MultipleLines', 'InternetService',  
              'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',  
              'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',  
              'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges',  
              'TotalCharges', 'Churn'], dtype=object)
```

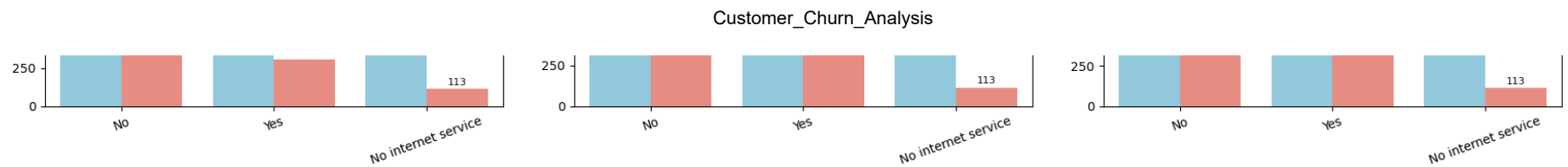
```
In [20]: # Columns to plot  
cols = [  
    'PhoneService', 'MultipleLines', 'InternetService',
```

```
'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',  
'TechSupport', 'StreamingTV', 'StreamingMovies'  
]  
  
# Create a 3x3 grid  
fig, axes = plt.subplots(3, 3, figsize=(18, 15))  
axes = axes.flatten()  
  
for i, col in enumerate(cols):  
    ax = sns.countplot(  
        data=df,  
        x=col,  
        hue='Churn',  
        ax=axes[i],  
        palette=["skyblue", "salmon"]  
    )  
  
    # Add data labels on every bar  
    for container in ax.containers:  
        ax.bar_label(  
            container,  
            fmt='%d',  
            fontsize=8,  
            padding=2  
        )  
  
    ax.set_title(f'Churn by {col}')  
    ax.set_xlabel('')  
    ax.tick_params(axis='x', rotation=20)  
  
    # Keep Legend only on the first subplot  
    if i != 0:  
        ax.legend_.remove()  
  
# Common Legend  
handles, labels = axes[0].get_legend_handles_labels()  
fig.legend(handles, labels, title='Churn',  
           loc='upper center', ncol=2)
```

```
plt.tight_layout(rect=[0, 0, 1, 0.95])  
plt.show()
```

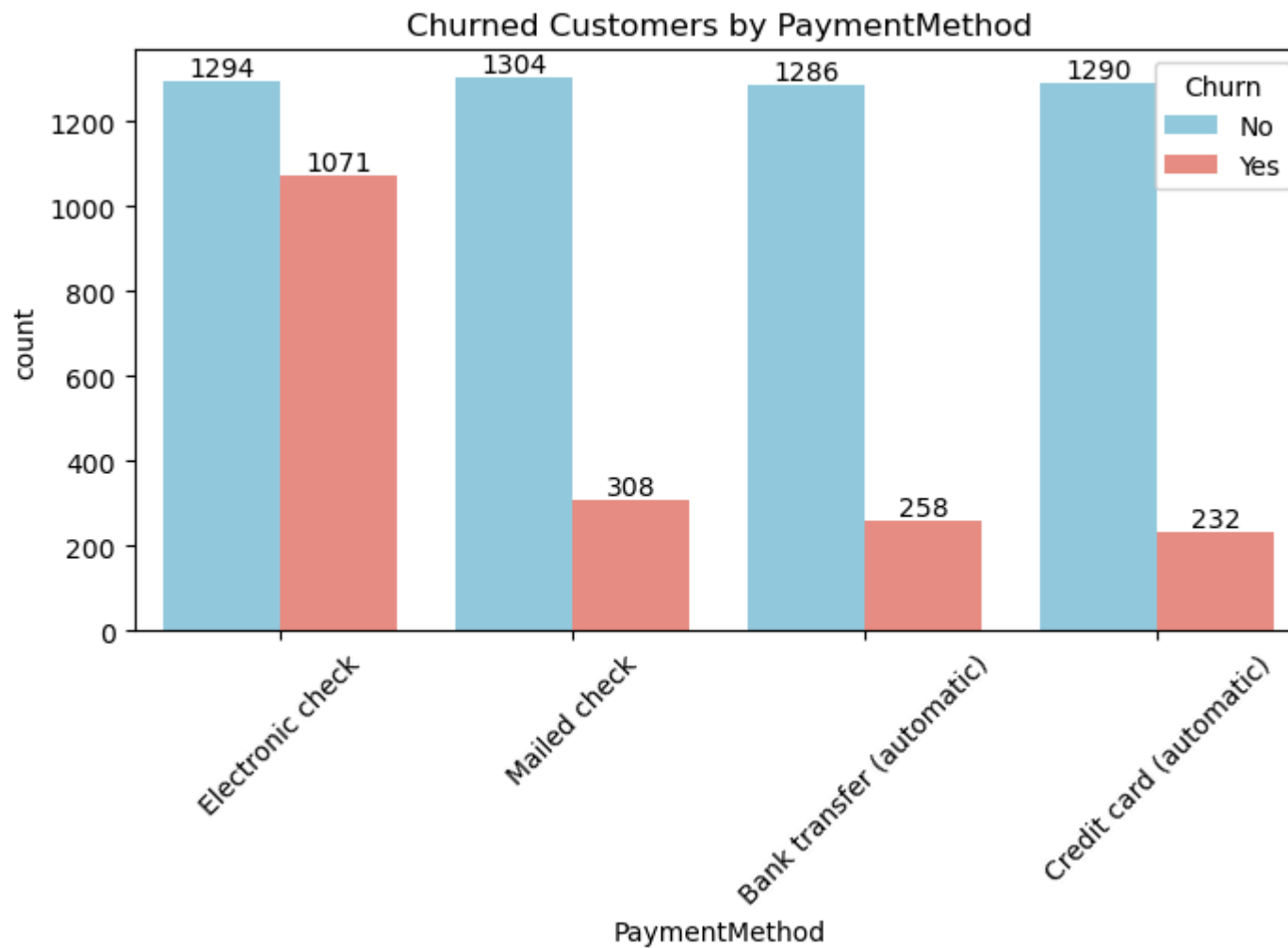
Customer_Churn_Analysis





Customers without value-added services like Online Security and Tech Support tend to churn more frequently.

```
In [21]: plt.figure(figsize=(8,4))
ax=sns.countplot(x="PaymentMethod",data=df,hue="Churn",palette=["skyblue", "salmon"])
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
plt.title("Churned Customers by PaymentMethod",fontsize=12)
plt.xticks(rotation=45)
plt.show()
```

Customers paying through electronic checks have the highest churn, while automatic payment methods show better customer retention.

In []: